ALEX LI

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EDUCATION

University of Washington, Seattle, WA — Expected June 2027
B.S., Computer Science

Relevant Coursework: Discrete Mathematics, Linear Algebra, Data Structures and Parallelism, Hardware/Software Interface, Systems Programming, Digital Circuit Design, Machine Learning, Distributed Systems, Data Management, Datacenter Systems

Awards & Achievements: Best Use of AI + Best Finance/Data Analytics Project (HackTech 2025), **1st Place** DubHacks 2023, **Futureforce Selectee**, **Y Combinator** (Winter 2026)

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, SQL, Java, JavaScript, SystemVerilog, HTML/CSS
- **Frameworks/Libraries:** Flask, Spring Boot, Express, React, Next, PyTorch, NumPy, pandas
- **Tools & Platforms:** Linux, Git, Make, PostgreSQL, Terraform, Docker, AWS, perf, FlameGraph, gRPC

RELEVANT EXPERIENCE

Oracle | *C, SQL, perf, FlameGraph* — June 2026 — September 2026
Software Engineering Intern - AI & Data Engines

- Designed and implemented a feature for vector indexes across DDL, DML, query execution, and index rebuilds to enable exact-match guarantees; profiled performance with **perf** and **FlameGraph** for an upcoming Oracle Database release.
- Designed and prototyped a hybrid LSH/IVF indexing and search algorithm, improving recall by **10% at equivalent query cost** and reducing wasted candidate comparisons by **33% at equivalent recall**.

Copperlane (YC W26) | *FastAPI, DynamoDB, S3* — November 2025 — May 2026
Co-founder & CTO: Nov 2025 - Feb 2026
Founding Engineer: March 2026 - May 2026

- Worked directly with early enterprise customers to identify operational bottlenecks, translate ambiguous workflows into product requirements, and rapidly ship solutions that helped grow the business to **five-figure contracted ARR**.
- Built an **SQS FIFO-based workflow and retry system** with ordered processing, deduplication, idempotent execution, and DLQs to make customer-facing workflows reliable under failures and redeliveries.

Chewy | *Spring Boot, Terraform, AWS, Mockito, Guice* — June 2025 — August 2025
Software Engineering Intern

- Built and deployed **backend microservices** for the Chewy Ads reporting platform, replacing a third-party scheduler with an **in-house job orchestration system** on AWS Lambda, SQS, and DynamoDB.
- Designed **retry-safe async workflows** with exponential backoff, dead-letter queues, and request deduplication.

UW Sensor Systems Lab | *FPGAs, Verilog, GTKWave, Python, Serial Interfaces* — December 2024 — June 2025
Undergraduate Research Assistant

- Developed FPGA drivers and a high-throughput UART control protocol, synchronizing 100+ ultrasonic transducers for acoustic-levitation.

PROJECTS

Othello Game Engine | *C++23, Multithreading, Profiling, gRPC, Docker* — GitHub

- Engineered a high-performance **parallel game engine** using iterative-deepening α-β search with principal variation search (PVS) and Zobrist-hashed transposition tables; optimized multithreaded throughput to 18.5M nodes/s by sharding the transposition table per thread and eliminating STL overhead in the move-generation hot path.

Multi-Paxos Key-Value Store | *Java, Multi-Paxos, Sharding*

- Designed and implemented a **fault-tolerant distributed key-value store** with leader election, replication, and dynamic shard rebalancing using the Multi-Paxos consensus algorithm.
- Built consensus-driven reconfiguration logic ensuring **linearizable consistency** under network partitions and crash faults; validated with adversarial distributed tests.